18020

B.C.A. Examination, May-2024

MATHEMATICS-III

(BCA-406)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt all the **five** questions. Each question carries three marks. $3\times5=15$

1. Explain periodic functions with examples.

2. Define Exact-differential equation. Check that the given differential equation is exact :

   $e^y dx+(xe^y+2y)dy=0$.

3. Find the general solution, of the following differential equation.

   $y''+y'-6y=0$

4. Define Polar form of complex number and find argument of $-\sqrt{3}-i$.

5. Find the magnitude of the gradient of the function $f=xyz^2$ at (1, 0, 2).

Section-B

(Short Answer Type Questions)

Note : Attempt any **two** questions out of the **three** questions. Each question carries 7.5 marks. $2\times7.5=15$

6. Explain Limit Comparison Test. Test the convergence of the following series.

   $\Sigma u_n = (n^3+1)^{1/3}-n$

7. Explain the monotonic sequence and Bounded Sequence. Show that

   $\lim \sqrt[n]{n} = 1$

8. (a) Compute real and imaginary part of

$$z = \frac{i-4}{2i-3}$$

(b) Compute Square root of

$z = -1-i$.

Section-C

(Descriptive Answer Type Questions)

Note : Attempt any **three** questions out of the following **five** questions. Each question carry equal marks.

$15 \times 3 = 45$

9. Find the Fourier series expansion for

$$f(x) = x + \frac{x^2}{4}, -\pi \le x \le \pi$$

10. (a) Solve :

$(x \log x)y' + y = 3x^3$

(b) Solve the following equation by finding an integrating factor

$(x^3 + xy^3)dx + 3y^2 dy = 0$.

P.T.O.

11. Find the general solution of the following equation :

$y'' - 3y' + 2y = 14 \sin 2x - 18 \cos 2x$.

12. Define the gradient and directional derivatives of a vector function. Find the directional derivative of

$f(x, y, z) = x^2 + y^2 + z^2$ at the point (1, 2, 3) in the direction of $3\hat{i} + 4\hat{j} + 10\hat{k}$.

13. (a) Explain Leibnitz test. Show that the series $\frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \ldots\ldots\ldots$ is conditionally convergent.

(b) Define absolute Convergent series. Show that the series

$$1 - \frac{1}{2^3} - \frac{1}{4^3} + \frac{1}{3^3} - \frac{1}{6^3} - \frac{1}{8^3} + \ldots\ldots$$

$$+ \frac{1}{(2n-1)^3} - \frac{1}{(4n-2)^3} - \frac{1}{(4n)^3} + \ldots\ldots$$

is absolutely convergent.

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BCA – IV Sem.

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18020

B.C.A. Examination, June–2023

MATHEMATICS-III

(BCA-406)

Time : 3 Hours] *[Maximum Marks : 75*

Note: Attempt all the Sections as per instructions.

Section–A

(Very Short Answer Type Questions)

Note : Attempt all the *five* questions. Each question carries 3 marks. 5 ×3=15

1. Define Fourier Series.

2. Solve : $(x+y)dx - (x-y)\,dy = 0$

3. Solve $y'' - 9y' + 20y = 0$

4. Find the argument of the following Complex number:

   $-1 - i\sqrt{3}$

5. If $f = x^2z\,\hat{i} - 2y^3z^2\,\hat{j} + xy^2z\,\hat{k}$ find div f at $(1, -1, 1)$.

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Section–B

(Short Answer Type Questions)

Note: Attempt any *two* questions out of the following three questions. Each question carries 7.5 marks. 2 × 7.5=15

6. Define Convergent sequence. Show that the sequence $\left\langle \frac{1}{n} \right\rangle$ has the limit 0.

7. Explain Cauchy's root test. Test for Convergence

   $$\sum\left(\frac{n+1}{n+2}\right)^n . x^n, \ (x > 0)$$

8. If Z_1 and Z_2 are complex numbers then prove that:

   $$|Z_1 + Z_2|^2 + |Z_1 - Z_2|^2 = \{|Z_1|^2 + |Z_2|^2\}$$

Section–C

(Descriptive Answer Questions)

Note: Attempt any *three* questions out of the following five questions. Each question carries 15 marks. 3 ×15=45

9. Find the Fourier series to represent $f(x) = \pi - x$, for $0 < x < 2\pi$.

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10. (a) Solve the following equations by finding an integrating factor :

$$xdy + ydx + 3x^3 y^4 dy = 0$$

(b) Solve :

$$xy^2 y' + y^3 = x \cos x;$$

11. Find the general solutions of the following equation:

$$y'' + 4y = 3 \sin x$$

12. Define curl and divergence of a vector. Prove the following vector identity:

$$\text{div}\left(\vec{u} \times \vec{v}\right) = \vec{v} \cdot \text{curl}\ \vec{u} - \vec{u} \cdot \text{curl}\ \vec{v}.$$

13. (a) Test the Convergence the series

$$x^2 + \frac{2^2}{3.4}x^4 + \frac{2^2.4^2}{3.4.5.6}x^6 + \frac{2^2.4^2.6^2}{3.4.5.6.7.8}x^8 + \text{——}$$

(v) Show that the series $\frac{2}{12} - \frac{3}{2^2} + \frac{4}{3^2} - \frac{5}{4^2} + \ldots$ Converges conditionally.

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A

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Roll No. ....

BCA - IV Sem.

18020

B.C.A. Examination, June-2022

MATHEMATICS-III

(BCA - 406)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

नोट : **सभी** खण्डों को निर्देशानुसार हल कीजिए।

Section - A/ खण्ड - क

(Very Short Answer Type Questions)

(अति लघु उत्तरीय प्रश्न)

Note : Attempt all the **five** questions. Each question carries 3 marks.

नोट : सभी **पाँच** प्रश्नों के उत्तर दीजिए। प्रत्येक प्रश्न 3 अंकों के है।

1. Show that

दिखाइये कि

$$|z_1+z_2|^2 + |z_1-z_2|^2 = 2|z|^2 + 2|z_2|^2$$

P.T.O.

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2. Define sequence with example.

अनुक्रम को उदाहरण सहित परिभाषित करो।

3. If $f(x, y, z) = 3x^2y - y^3z^2$, then find grad f at point (1, -2, -1).

यदि $f(x, y, z) = 3x^2y - y^3z^2$ तब ग्रेड f का मान बिन्दु (1, -2, -1) पर बताइये।

4. Solve

हल करो

$$\frac{dy}{dx} = \sin(x+y)$$

5. Solve

हल करो

$$(D^2+3D^2+3D+1)y = e^x + e^{-x}$$

Section - B / खण्ड - ख

(Short Answer Type Questions)

(लघु उत्तरीय प्रश्न)

Note : Attempt any **two** questions. Each question carries 7.5 marks.

नोट : किन्हीं **दो** प्रश्नों के उत्तर दीजिए। प्रत्येक प्रश्न 7.5 अंकों के हैं।

6. Determine the regions defined by

क्षेत्र का निर्धारण कीजिए, यदि

$$|z-1| + |z+1| \leqslant 4$$

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7. Solve

हल करो

$$xdx + ydy + \frac{xdy - ydx}{x^2 + y^2} = 0$$

8. Show

दिखाइये

$$\nabla r^{-3} = -3r^{-5}\vec{r}, \text{ where } \vec{r} = xi + yj + z\hat{k}$$

Section - C / खण्ड - ग

(Detailed Answer Type Questions)

(विस्तृत उत्तरीय प्रश्न)

Note : Attempt any **three** questions. Each question carries 15 marks.

नोट : किन्हीं **तीन** प्रश्नों के उत्तर दीजिये। प्रत्येक प्रश्न 15 अंकों के हैं।

9. Show that

दिखाइये कि

$$\lim_{n \to \infty}\left(1 + \frac{1}{n}\right)^n = e, \text{ where } 2 < e < 3$$

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10. Test the Convergence of following series

निम्न श्रेणी की अभिसारिता की जांच करो

$$\frac{1}{4} + \frac{9}{2^2.4^2}x + \frac{3^2.5^2}{2^2.4^2.6^2}x^2 + \frac{3^2.5^2.7^2}{2^2.4^2.6^2.8^2}x^3 + \ldots..$$

11. Obtain the Fourier series of $f(x) = \left(\frac{\pi - x}{2}\right)$ in the interval $(0, 2\pi)$ and hence deduce

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots..$$

अन्तराल $(0, 2\pi)$ के लिए $f(x) = \left(\frac{\pi - x}{2}\right)$ को फोरियर श्रेणी में दिखाओ और दिखाओं कि

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots..$$

12. Solve

हल करो

$$(1+y^2)dx-(\tan^{-1}y-x)dy = 0$$

13. Solve

हल करो

$$(D^2+1)y = \sin x \sin 2x$$

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Total Questions : 13] [Printed Pages : 4

18020

B.C.A. IVth Semester Examination, May-2019

MATHEMATICS-III

(BCA-406)

Time : 3 Hrs.] *[M.M. : 75*

Note :– Attempt all the Sections as per instructions.

Section–A

(Very Short Answer Type Questions)

Note :– Attempt all the *five* questions. Each question carries 3 marks.

1. Find the order and degree of the differential equation :

$$\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^4 + y = 0$$

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2. State Leibnitz test.

Or

Cauchy's root test.

3. Express $1 - i$ in the modulus amplitude form.

4. If $r = \sin t\,\hat{i} + \cos t\,\hat{j} + \hat{k}$, find :

(i) $\frac{dr}{dt}$

(ii) $\left|\frac{dr}{dt}\right|$

5. Solve :

$$\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 12y = 0$$

Section–B

(Short Answer Type Questions)

Note :– Attempt any *two* questions out of the following three questions. Each question carries 7½ marks.

6. If z_1 and z_2 are any complex numbers, then prove that :

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\{|z_1|^2 + |z_2|^2\}$$

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7. Solve :

$$(1 + x^2)dy = (1 + y^2)dx$$

8. Prove that :

$$\lim \frac{1}{n}\left(1 + 2^{1/2} + 3^{1/3} + \ldots\ldots + n^{1/n}\right) = 1$$

Section–C

(Long Answer Type Questions)

Note :– Attempt any *three* questions out of the following five questions. Each question carries 15 marks.

9. (a) If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, then prove that :

$$x + y + z = xyz$$

(b) Test the convergence of the series whose nth term is :

$$\sqrt{n+1} - \sqrt{n}$$

10. Find the directional derivative of $f(x, y, z) = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$.

11. Find the Fourier series for the function $f(x) = |x|$, $-\pi < x < \pi$. Hence deduce that :

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots\ldots\ldots$$

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12. (a) Solve :

$$\frac{d^2y}{dx^2} + 9y = \cos 2x + \sin 2x$$

(b) Solve :

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = e^{-3x}$$

13. (a) Solve :

$$(e^y + 1)\cos x\, dx + e^y \sin x\, dy = 0$$

(b) Solve :

$$(x + 2y^3)\frac{dy}{dx} = y$$

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(20518) Roll No.

BCA- IV Sem.

18020

B. C. A. Examination, May 2018

MATHEMATICS-III

(BCA-406)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required.

3×5=15

1. Express $\frac{2+3i}{4+5i}$ in the form of $x+iy$. 3

2. If $x+iy=\frac{3}{2+\cos\theta+i\sin\theta}$, prove that : 3

$$(x-1)(x-3)+y^2=0.$$

3. Test for convergence $\sum_{n=1}^{\infty}\sin\frac{1}{n}$. 3

4. If $\vec{r}=a\hat{i}+b\hat{j}+c\hat{k}$, find the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r}$. 3

5. For any scalar function $\phi(x, y, z)$, prove that curl grad $\phi=0$. 3

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. Solve $(x+1)\frac{dy}{dx}=x(y^2+1)$. 7½

7. If $\tan^{-1} a + \tan^{-1} b + \tan^{-1} c = \pi$, then prove that $a + b + c = abc$. 7½

8. Solve $\frac{dy}{dx} = \frac{1+y^2}{\tan^{-1} y - x}$. 7½

Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. (a) Test the convergence of the series : 7½

$$\frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4}\right)^2 x^2 + \left(\frac{4}{5}\right)^3 x^3 + \ldots\ldots$$

(b) Test for convergence the series whose nth term is $\frac{r^n}{n^n}$, $r > 0$. 7½

10. (a) Show that the vector field defined by :

$$\vec{F} = 2xyz^3\hat{i} + x^2z^3\hat{j} + 3x^2yz^2\hat{k}$$

is irrotational. Find the scalar potential u such that $\vec{F} = \text{grad } u = \nabla u$. 7½

(b) If $\vec{E}$ and $\vec{H}$ are irrotational, prove that $\vec{E} \times \vec{H}$ is solenoidal. https://www.ccsustudy.com 7½

11. (a) Test for the convergence of the series

$$\sum_{n=2}^{\infty} \frac{1}{(\log n)^n}.$$ 7½

(b) Find the directional derivative of $f(x, y, z) = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$. 7½

12. (a) Solve $y\sec^2 x + (y+7)\tan x \frac{dy}{dx} = 0$. 7½

(b) Solve $(1+x^2)\frac{dy}{dx} + 2xy = \cos x$. 7½

13. (a) Find the directional derivative of $f(x, y, z) = x^2y^2z^2$ at the point $(1, 1, -1)$ in the direction of the tangent to the curve $x = e^t$, $y = 2\sin t$, $z = t - \cos t$, at $t = 0$. 7½

(b) Solve $x(e^y + 4)dx + e^{x+y}dy = 0$. 7½

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